Does childhood meat eating contribute to sex differences in risk factors for ischaemic heart disease in a developing population?

BACKGROUND: A male epidemic of ischaemic heart disease (IHD) emerges with economic development. It has previously been hypothesised that this epidemic is due to nutritionally driven levels of pubertal sex steroids, which lead to a more atherogenic body shape and lipid profile in boys but not girls, without any sex-specific effects on glucose metabolism. This study tests this hypothesis by examining the association of childhood meat eating with IHD risk in a developing Chinese population. METHODS: Multivariable linear and censored regression was used in a cross-sectional study of 19,418 Chinese older (≥ 50 years) men and women from the Guangzhou Biobank Cohort Study (phases 2 and 3) to assess the adjusted associations of childhood meat eating with waist to hip ratio (WHR), high-density lipoprotein cholesterol and fasting plasma glucose. RESULTS: Adjusted for age, childhood hunger, life-course socioeconomic position and current lifestyle childhood almost daily meat eating compared with less than weekly meat eating was associated with higher WHR (0.007, 95% CI 0.0003 to 0.01) in men but not women. No association with fasting glucose was observed. CONCLUSIONS: Given the potential limitations of this study, especially the crude nature of the exposure and modest findings, the results should be considered as preliminary. However, they do lend support to the hypothesis that the male epidemic of premature IHD and sexual divergence in IHD rates that occur with economic development may be nutritionally driven in childhood. In elucidating the developmental origins of non-communicable chronic diseases, more attention should be focused on the sociohistorical context and the role of puberty.